Randy Ren

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Experience

Technical Project Coordinator January 2026 – August 2026
SAP — Internship *Vancouver, BC*

- Built an **open-source context-store tool** into the team's agentic workflow to stop engineers from re-explaining lost context: it indexes each session for hybrid keyword and vector search (**RAG**) and re-injects relevant history automatically, improving token efficiency by **23%**. Extended it into a **Git-backed team space** where engineers publish and search one another's context.
- Shipped an agentic triage system for the open-source-security backlog that buckets overdue tickets, routes each to its current owner, and drafts the **Jira**/Slack/Outlook follow-up; published it on SAP's internal skills marketplace for **300+** developers.
- Coordinated release operations for the SAP Analytics Cloud Vancouver org, working across senior managers, engineers, and architects: audited closed release tickets against semi-annual audit standards and translated release-approval decks into requirements-tracked **Jira** sub-tasks.

Machine Learning Engineer September 2025 – December 2025
Ledcor — Internship *Vancouver, BC*

- Productized the fiber-installation audit workflow from the prior Ledcor Project Manager term: a **YOLO** detector proposes candidate defects and a vision-language model adjudicates each against the installation spec, automating **10,000** QA inspections and saving around **330** hours of manual review annually.
- Fine-tuned the detector and evaluated the full pipeline on **60,000** historical QA decisions spanning ten years, reaching **99%** agreement with human QA coordinators across a **12,000**-case evaluation dataset.

Project Manager May 2025 – August 2025
Ledcor — Internship *Vancouver, BC*

- Ran **QA** test scheduling and field-inspection tracking for a multi-site fiber build program, owning the cross-functional cadence across field superintendents, design engineers, and external build partners.
- Owned the escalation path when field findings conflicted with design intent, prioritizing each conflict and driving it to a documented decision.

Software Engineer May 2024 – August 2024
Visa — Internship *San Francisco, CA*

- Built and evaluated personalized authorization models (**Python**, **scikit-learn**) for payment anomaly detection in Visa's **Smarter Stand-In Processing (STIP)** path, the fallback that authorizes transactions when an issuer is unreachable, and presented the evaluation to the owning team.

Projects

Umbra | Headless WebKit browser for AI agents | github.com/randyren278/umbra

- Built a headless browser for AI agents in **Swift/WebKit**, with persistent sign-ins and **CLI/MCP** interfaces, while using **50% fewer tokens** than Playwright MCP in local observation benchmarks.
- Achieved **3.8× faster snapshots** and **10.5× faster clicks** than Playwright Chromium across 20-run local benchmarks using a persistent daemon socket.

Hephaestus | Evidence-first evaluation control plane for AI agents | github.com/randyren278/hephaestus

- Built an eight-crate **Rust** control plane that A/B-tests AI agents: each version runs the same tasks in its own macOS **Seatbelt** sandbox, scored by a sealed evaluator it cannot read, under correctness, cost, and latency gates.

Straits | Maritime oil-chokepoint intelligence dashboard | straits.randyren.org

- Built a live dashboard that scores every ship at four Middle East oil chokepoints 0–100 for risk and anomalies and predicts where moving ships are between reports, using a **Kalman** filter and a walk-forward learner choosing among six estimators.

Technical Skills

AI & Agents: LLM agents, agent harnesses, human-in-the-loop supervision, Claude Code skills and hooks, MCP servers, RAG (FTS5 + sqlite-vec, rank fusion), evals/benchmarking, scikit-learn, YOLO
Product & Project Management: Jira, Confluence, product roadmaps, backlog prioritization, cross-functional stakeholder management, release management, SDLC, QA planning, requirements, technical specs
Languages & Tools: Python, TypeScript, C/C++, SQL, Rust, Swift, MATLAB, Bash; Git, Docker, CI, Playwright, Next.js/React, PostgreSQL

Education

University of British Columbia Vancouver, BC
Bachelor of Applied Science (B.A.Sc.), Electrical Engineering *September 2023 – April 2028*

Activities

UBC Thunderbirds Varsity Track & Field (U SPORTS/NAIA) | 800m PB 1:54.79, 5000m PB 15:17